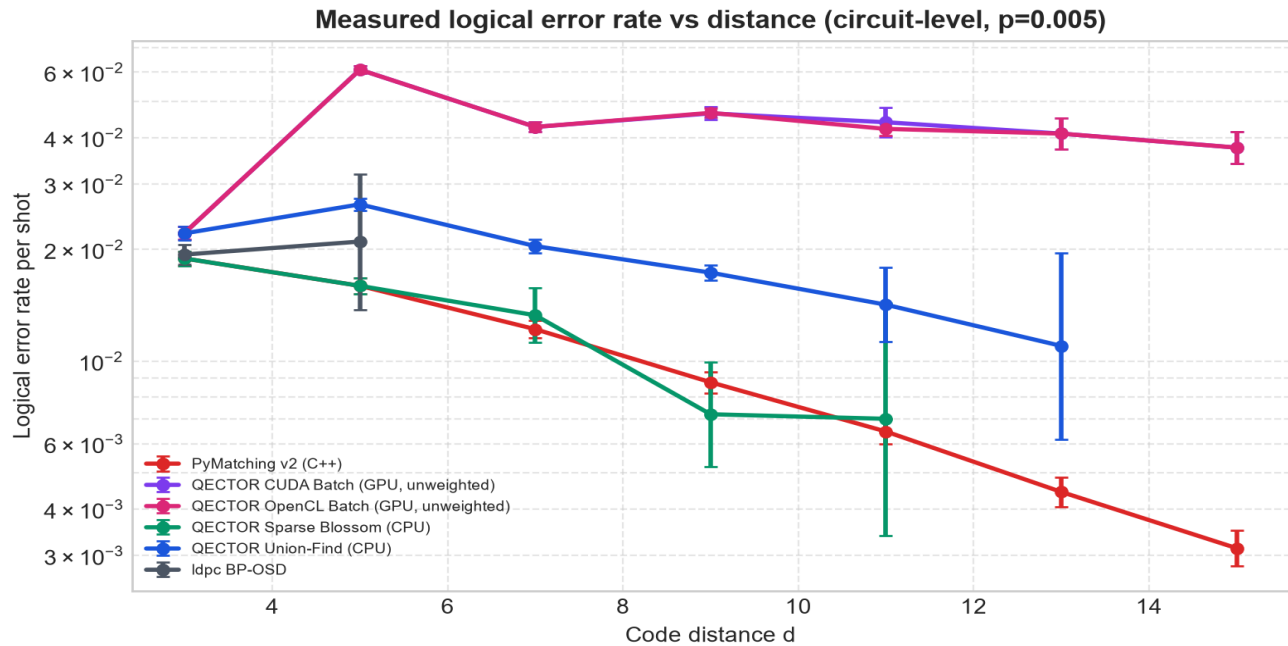
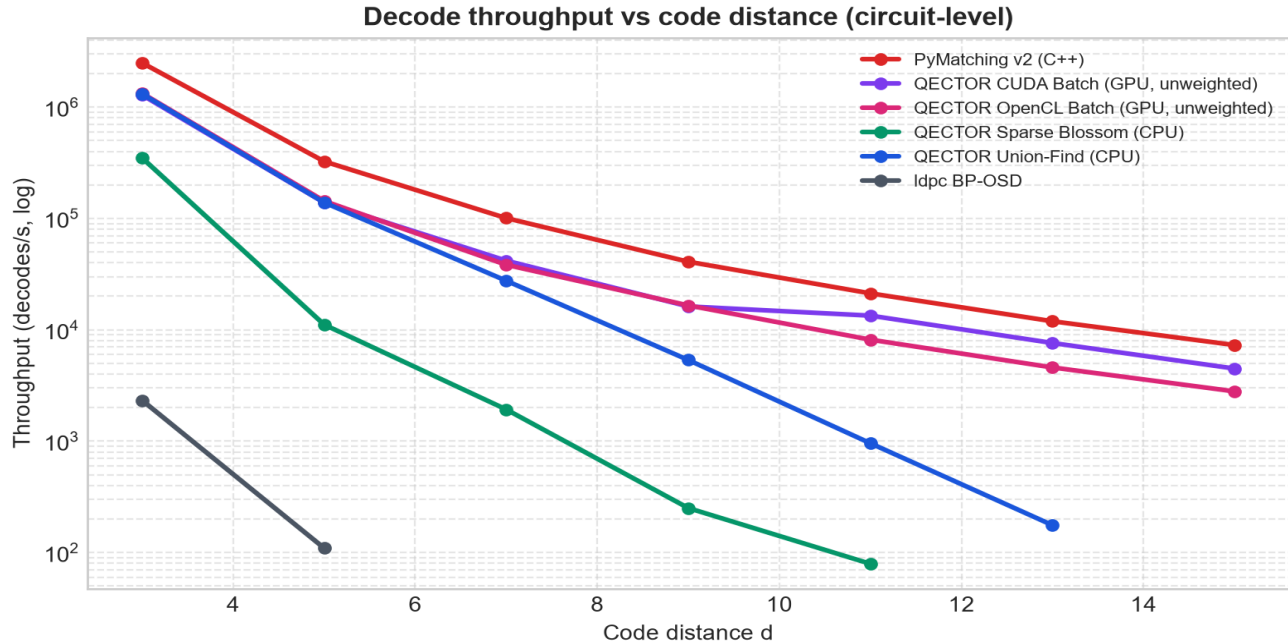


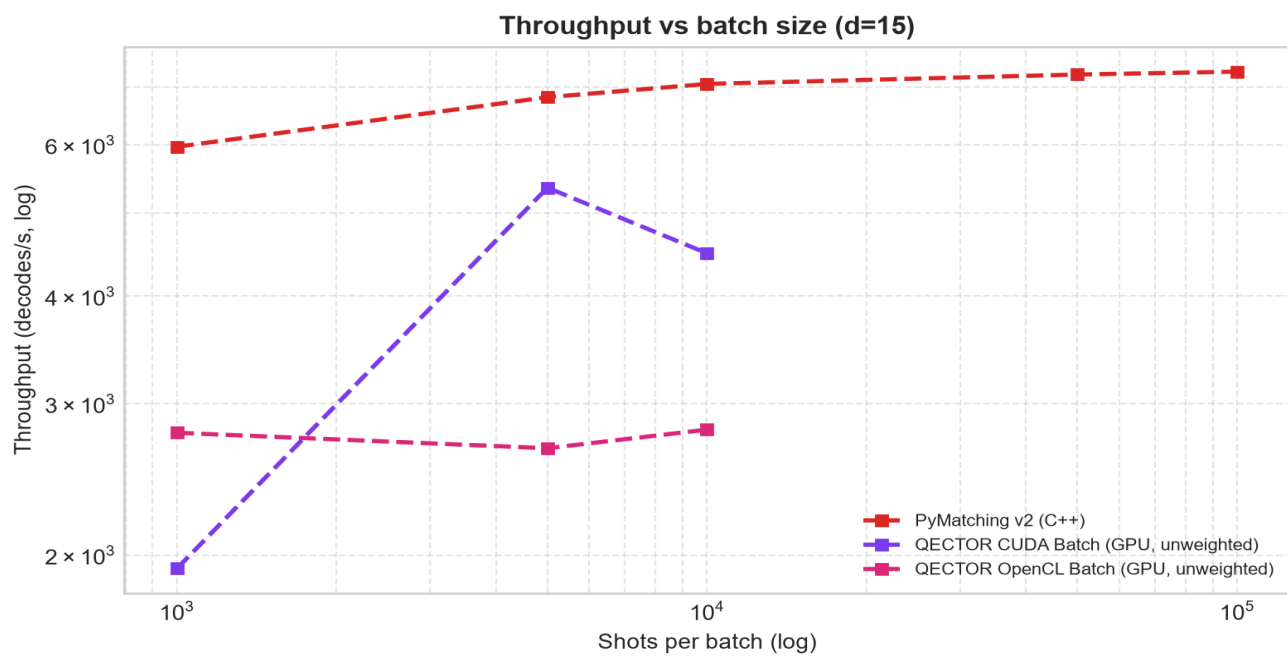
# QECTOR v0.7.0 — circuit-level decoder comparison

Measured with `ler.estimate_ler_circuit_level`: identical circuit, DEM, samples and decode\_batch resolver for every decoder, scored against the circuit's logical observables. Validated by `ler.assert_comparable`. No extrapolated cells.

Noise model: circuit-level,  $p=0.005$  | Commit: b057aeee34 | Generated: 2026-08-01T03:01:35Z

Platform: Windows-10-10.0.26100-SP0 | Python: 3.11.9





## Measured results (137 cells)

| Decoder                               | d | Shots   | Errors | LER            | 95% CI           | dec/s            | vs PM  |
|---------------------------------------|---|---------|--------|----------------|------------------|------------------|--------|
| PyMatching v2 (C++)                   | 3 | 1,000   | 13     | <b>0.01300</b> | [0.0076, 0.0221] | <b>1,949,698</b> | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 3 | 1,000   | 16     | <b>0.01600</b> | [0.0099, 0.0258] | <b>308,604</b>   | 0.16x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 3 | 1,000   | 16     | <b>0.01600</b> | [0.0099, 0.0258] | <b>359,790</b>   | 0.18x  |
| QECTOR Sparse Blossom (CPU)           | 3 | 1,000   | 13     | <b>0.01300</b> | [0.0076, 0.0221] | <b>250,645</b>   | 0.13x  |
| QECTOR Union-Find (CPU)               | 3 | 1,000   | 16     | <b>0.01600</b> | [0.0099, 0.0258] | <b>382,614</b>   | 0.20x  |
| ldpc BP-OSD                           | 3 | 1,000   | 12     | <b>0.01200</b> | [0.0069, 0.0209] | <b>2,223</b>     | 0.001x |
| PyMatching v2 (C++)                   | 3 | 5,000   | 84     | <b>0.01680</b> | [0.0136, 0.0208] | <b>2,394,865</b> | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 3 | 5,000   | 105    | <b>0.02100</b> | [0.0174, 0.0254] | <b>1,572,772</b> | 0.66x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 3 | 5,000   | 105    | <b>0.02100</b> | [0.0174, 0.0254] | <b>1,311,647</b> | 0.55x  |
| QECTOR Sparse Blossom (CPU)           | 3 | 5,000   | 84     | <b>0.01680</b> | [0.0136, 0.0208] | <b>319,211</b>   | 0.13x  |
| QECTOR Union-Find (CPU)               | 3 | 5,000   | 105    | <b>0.02100</b> | [0.0174, 0.0254] | <b>643,227</b>   | 0.27x  |
| ldpc BP-OSD                           | 3 | 5,000   | 87     | <b>0.01740</b> | [0.0141, 0.0214] | <b>2,345</b>     | 0.001x |
| PyMatching v2 (C++)                   | 3 | 10,000  | 202    | <b>0.02020</b> | [0.0176, 0.0231] | <b>2,433,682</b> | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 3 | 10,000  | 235    | <b>0.02350</b> | [0.0207, 0.0267] | <b>1,700,044</b> | 0.70x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 3 | 10,000  | 235    | <b>0.02350</b> | [0.0207, 0.0267] | <b>1,557,681</b> | 0.64x  |
| QECTOR Sparse Blossom (CPU)           | 3 | 10,000  | 202    | <b>0.02020</b> | [0.0176, 0.0231] | <b>339,989</b>   | 0.14x  |
| QECTOR Union-Find (CPU)               | 3 | 10,000  | 233    | <b>0.02330</b> | [0.0205, 0.0264] | <b>835,492</b>   | 0.34x  |
| ldpc BP-OSD                           | 3 | 10,000  | 199    | <b>0.01990</b> | [0.0173, 0.0228] | <b>2,332</b>     | 0.001x |
| PyMatching v2 (C++)                   | 3 | 50,000  | 960    | <b>0.01920</b> | [0.0180, 0.0204] | <b>2,513,232</b> | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 3 | 50,000  | 1131   | <b>0.02262</b> | [0.0214, 0.0240] | <b>1,121,034</b> | 0.45x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 3 | 50,000  | 1131   | <b>0.02262</b> | [0.0214, 0.0240] | <b>1,239,695</b> | 0.49x  |
| QECTOR Sparse Blossom (CPU)           | 3 | 50,000  | 960    | <b>0.01920</b> | [0.0180, 0.0204] | <b>345,329</b>   | 0.14x  |
| QECTOR Union-Find (CPU)               | 3 | 50,000  | 1128   | <b>0.02256</b> | [0.0213, 0.0239] | <b>1,188,436</b> | 0.47x  |
| ldpc BP-OSD                           | 3 | 50,000  | 969    | <b>0.01938</b> | [0.0182, 0.0206] | <b>2,327</b>     | 0.001x |
| PyMatching v2 (C++)                   | 3 | 100,000 | 1891   | <b>0.01891</b> | [0.0181, 0.0198] | <b>2,497,016</b> | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 3 | 100,000 | 2215   | <b>0.02215</b> | [0.0213, 0.0231] | <b>1,285,174</b> | 0.52x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 3 | 100,000 | 2215   | <b>0.02215</b> | [0.0213, 0.0231] | <b>1,331,565</b> | 0.53x  |
| QECTOR Sparse Blossom (CPU)           | 3 | 100,000 | 1891   | <b>0.01891</b> | [0.0181, 0.0198] | <b>349,901</b>   | 0.14x  |
| QECTOR Union-Find (CPU)               | 3 | 100,000 | 2210   | <b>0.02210</b> | [0.0212, 0.0230] | <b>1,314,712</b> | 0.53x  |
| PyMatching v2 (C++)                   | 5 | 1,000   | 19     | <b>0.01900</b> | [0.0122, 0.0295] | <b>286,886</b>   | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 5 | 1,000   | 60     | <b>0.06000</b> | [0.0469, 0.0765] | <b>86,090</b>    | 0.30x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 5 | 1,000   | 60     | <b>0.06000</b> | [0.0469, 0.0765] | <b>126,542</b>   | 0.44x  |
| QECTOR Sparse Blossom (CPU)           | 5 | 1,000   | 19     | <b>0.01900</b> | [0.0122, 0.0295] | <b>11,669</b>    | 0.04x  |
| QECTOR Union-Find (CPU)               | 5 | 1,000   | 26     | <b>0.02600</b> | [0.0178, 0.0378] | <b>95,464</b>    | 0.33x  |
| ldpc BP-OSD                           | 5 | 1,000   | 21     | <b>0.02100</b> | [0.0138, 0.0319] | <b>109</b>       | 0x     |
| PyMatching v2 (C++)                   | 5 | 5,000   | 73     | <b>0.01460</b> | [0.0116, 0.0183] | <b>314,725</b>   | 1.00x  |

| Decoder                               | d | Shots   | Errors | LER            | 95% CI           | dec/s          | vs PM |
|---------------------------------------|---|---------|--------|----------------|------------------|----------------|-------|
| QECTOR CUDA Batch (GPU, unweighted)   | 5 | 5,000   | 336    | <b>0.06720</b> | [0.0606, 0.0745] | <b>230,917</b> | 0.73x |
| QECTOR OpenCL Batch (GPU, unweighted) | 5 | 5,000   | 336    | <b>0.06720</b> | [0.0606, 0.0745] | <b>231,073</b> | 0.73x |
| QECTOR Sparse Blossom (CPU)           | 5 | 5,000   | 73     | <b>0.01460</b> | [0.0116, 0.0183] | <b>11,511</b>  | 0.04x |
| QECTOR Union-Find (CPU)               | 5 | 5,000   | 148    | <b>0.02960</b> | [0.0253, 0.0347] | <b>118,257</b> | 0.38x |
| PyMatching v2 (C++)                   | 5 | 10,000  | 166    | <b>0.01660</b> | [0.0143, 0.0193] | <b>322,491</b> | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 5 | 10,000  | 600    | <b>0.06000</b> | [0.0555, 0.0648] | <b>243,179</b> | 0.75x |
| QECTOR OpenCL Batch (GPU, unweighted) | 5 | 10,000  | 600    | <b>0.06000</b> | [0.0555, 0.0648] | <b>172,102</b> | 0.53x |
| QECTOR Sparse Blossom (CPU)           | 5 | 10,000  | 166    | <b>0.01660</b> | [0.0143, 0.0193] | <b>11,415</b>  | 0.04x |
| QECTOR Union-Find (CPU)               | 5 | 10,000  | 268    | <b>0.02680</b> | [0.0238, 0.0302] | <b>130,041</b> | 0.40x |
| PyMatching v2 (C++)                   | 5 | 50,000  | 809    | <b>0.01618</b> | [0.0151, 0.0173] | <b>327,439</b> | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 5 | 50,000  | 3041   | <b>0.06082</b> | [0.0588, 0.0629] | <b>137,869</b> | 0.42x |
| QECTOR OpenCL Batch (GPU, unweighted) | 5 | 50,000  | 3041   | <b>0.06082</b> | [0.0588, 0.0629] | <b>141,549</b> | 0.43x |
| QECTOR Sparse Blossom (CPU)           | 5 | 50,000  | 810    | <b>0.01620</b> | [0.0151, 0.0173] | <b>11,101</b>  | 0.03x |
| QECTOR Union-Find (CPU)               | 5 | 50,000  | 1356   | <b>0.02712</b> | [0.0257, 0.0286] | <b>136,080</b> | 0.42x |
| PyMatching v2 (C++)                   | 5 | 100,000 | 1596   | <b>0.01596</b> | [0.0152, 0.0168] | <b>325,421</b> | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 5 | 100,000 | 6094   | <b>0.06094</b> | [0.0595, 0.0624] | <b>138,721</b> | 0.43x |
| QECTOR OpenCL Batch (GPU, unweighted) | 5 | 100,000 | 6094   | <b>0.06094</b> | [0.0595, 0.0624] | <b>143,445</b> | 0.44x |
| QECTOR Sparse Blossom (CPU)           | 5 | 100,000 | 1596   | <b>0.01596</b> | [0.0152, 0.0168] | <b>11,121</b>  | 0.03x |
| QECTOR Union-Find (CPU)               | 5 | 100,000 | 2645   | <b>0.02645</b> | [0.0255, 0.0275] | <b>139,133</b> | 0.43x |
| PyMatching v2 (C++)                   | 7 | 1,000   | 11     | <b>0.01100</b> | [0.0062, 0.0196] | <b>90,436</b>  | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 7 | 1,000   | 44     | <b>0.04400</b> | [0.0329, 0.0586] | <b>21,782</b>  | 0.24x |
| QECTOR OpenCL Batch (GPU, unweighted) | 7 | 1,000   | 44     | <b>0.04400</b> | [0.0329, 0.0586] | <b>39,890</b>  | 0.44x |
| QECTOR Sparse Blossom (CPU)           | 7 | 1,000   | 11     | <b>0.01100</b> | [0.0062, 0.0196] | <b>1,917</b>   | 0.02x |
| QECTOR Union-Find (CPU)               | 7 | 1,000   | 18     | <b>0.01800</b> | [0.0114, 0.0283] | <b>11,689</b>  | 0.13x |
| PyMatching v2 (C++)                   | 7 | 5,000   | 74     | <b>0.01480</b> | [0.0118, 0.0185] | <b>96,754</b>  | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 7 | 5,000   | 218    | <b>0.04360</b> | [0.0383, 0.0496] | <b>69,309</b>  | 0.72x |
| QECTOR OpenCL Batch (GPU, unweighted) | 7 | 5,000   | 218    | <b>0.04360</b> | [0.0383, 0.0496] | <b>62,729</b>  | 0.65x |
| QECTOR Sparse Blossom (CPU)           | 7 | 5,000   | 75     | <b>0.01500</b> | [0.0120, 0.0188] | <b>1,870</b>   | 0.02x |
| QECTOR Union-Find (CPU)               | 7 | 5,000   | 107    | <b>0.02140</b> | [0.0177, 0.0258] | <b>23,848</b>  | 0.25x |
| PyMatching v2 (C++)                   | 7 | 10,000  | 130    | <b>0.01300</b> | [0.0110, 0.0154] | <b>99,588</b>  | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 7 | 10,000  | 409    | <b>0.04090</b> | [0.0372, 0.0450] | <b>63,533</b>  | 0.64x |
| QECTOR OpenCL Batch (GPU, unweighted) | 7 | 10,000  | 409    | <b>0.04090</b> | [0.0372, 0.0450] | <b>52,989</b>  | 0.53x |
| QECTOR Sparse Blossom (CPU)           | 7 | 10,000  | 133    | <b>0.01330</b> | [0.0112, 0.0157] | <b>1,925</b>   | 0.02x |
| QECTOR Union-Find (CPU)               | 7 | 10,000  | 209    | <b>0.02090</b> | [0.0183, 0.0239] | <b>25,554</b>  | 0.26x |
| PyMatching v2 (C++)                   | 7 | 50,000  | 619    | <b>0.01238</b> | [0.0114, 0.0134] | <b>98,837</b>  | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 7 | 50,000  | 2197   | <b>0.04394</b> | [0.0422, 0.0458] | <b>40,703</b>  | 0.41x |

| Decoder                               | d  | Shots   | Errors | LER            | 95% CI           | dec/s          | vs PM  |
|---------------------------------------|----|---------|--------|----------------|------------------|----------------|--------|
| QECTOR OpenCL Batch (GPU, unweighted) | 7  | 50,000  | 2197   | <b>0.04394</b> | [0.0422, 0.0458] | <b>37,017</b>  | 0.38x  |
| QECTOR Union-Find (CPU)               | 7  | 50,000  | 1055   | <b>0.02110</b> | [0.0199, 0.0224] | <b>24,487</b>  | 0.25x  |
| PyMatching v2 (C++)                   | 7  | 100,000 | 1220   | <b>0.01220</b> | [0.0115, 0.0129] | <b>100,735</b> | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 7  | 100,000 | 4274   | <b>0.04274</b> | [0.0415, 0.0440] | <b>41,777</b>  | 0.41x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 7  | 100,000 | 4274   | <b>0.04274</b> | [0.0415, 0.0440] | <b>38,331</b>  | 0.38x  |
| QECTOR Union-Find (CPU)               | 7  | 100,000 | 2042   | <b>0.02042</b> | [0.0196, 0.0213] | <b>27,571</b>  | 0.27x  |
| PyMatching v2 (C++)                   | 9  | 1,000   | 7      | <b>0.00700</b> | [0.0034, 0.0144] | <b>36,641</b>  | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 9  | 1,000   | 56     | <b>0.05600</b> | [0.0434, 0.0720] | <b>9,527</b>   | 0.26x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 9  | 1,000   | 56     | <b>0.05600</b> | [0.0434, 0.0720] | <b>16,533</b>  | 0.45x  |
| QECTOR Sparse Blossom (CPU)           | 9  | 1,000   | 7      | <b>0.00700</b> | [0.0034, 0.0144] | <b>248</b>     | 0.007x |
| QECTOR Union-Find (CPU)               | 9  | 1,000   | 14     | <b>0.01400</b> | [0.0084, 0.0234] | <b>1,852</b>   | 0.05x  |
| PyMatching v2 (C++)                   | 9  | 5,000   | 36     | <b>0.00720</b> | [0.0052, 0.0100] | <b>40,426</b>  | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 9  | 5,000   | 224    | <b>0.04480</b> | [0.0394, 0.0509] | <b>31,021</b>  | 0.77x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 9  | 5,000   | 224    | <b>0.04480</b> | [0.0394, 0.0509] | <b>25,001</b>  | 0.62x  |
| QECTOR Sparse Blossom (CPU)           | 9  | 5,000   | 36     | <b>0.00720</b> | [0.0052, 0.0100] | <b>251</b>     | 0.006x |
| QECTOR Union-Find (CPU)               | 9  | 5,000   | 85     | <b>0.01700</b> | [0.0138, 0.0210] | <b>6,285</b>   | 0.15x  |
| PyMatching v2 (C++)                   | 9  | 10,000  | 78     | <b>0.00780</b> | [0.0063, 0.0097] | <b>39,081</b>  | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 9  | 10,000  | 431    | <b>0.04310</b> | [0.0393, 0.0473] | <b>27,047</b>  | 0.69x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 9  | 10,000  | 431    | <b>0.04310</b> | [0.0393, 0.0473] | <b>21,219</b>  | 0.54x  |
| QECTOR Union-Find (CPU)               | 9  | 10,000  | 156    | <b>0.01560</b> | [0.0134, 0.0182] | <b>5,077</b>   | 0.13x  |
| PyMatching v2 (C++)                   | 9  | 50,000  | 450    | <b>0.00900</b> | [0.0082, 0.0099] | <b>40,959</b>  | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 9  | 50,000  | 2324   | <b>0.04648</b> | [0.0447, 0.0484] | <b>16,217</b>  | 0.40x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 9  | 50,000  | 2324   | <b>0.04648</b> | [0.0447, 0.0484] | <b>16,127</b>  | 0.39x  |
| QECTOR Union-Find (CPU)               | 9  | 50,000  | 827    | <b>0.01654</b> | [0.0155, 0.0177] | <b>5,361</b>   | 0.13x  |
| PyMatching v2 (C++)                   | 9  | 100,000 | 878    | <b>0.00878</b> | [0.0082, 0.0094] | <b>40,874</b>  | 1.00x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 9  | 100,000 | 4663   | <b>0.04663</b> | [0.0453, 0.0480] | <b>16,555</b>  | 0.41x  |
| QECTOR Union-Find (CPU)               | 9  | 100,000 | 1732   | <b>0.01732</b> | [0.0165, 0.0181] | <b>5,354</b>   | 0.13x  |
| PyMatching v2 (C++)                   | 11 | 1,000   | 7      | <b>0.00700</b> | [0.0034, 0.0144] | <b>18,455</b>  | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 11 | 1,000   | 44     | <b>0.04400</b> | [0.0329, 0.0586] | <b>5,663</b>   | 0.31x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 11 | 1,000   | 44     | <b>0.04400</b> | [0.0329, 0.0586] | <b>7,898</b>   | 0.43x  |
| QECTOR Sparse Blossom (CPU)           | 11 | 1,000   | 7      | <b>0.00700</b> | [0.0034, 0.0144] | <b>79</b>      | 0.004x |
| QECTOR Union-Find (CPU)               | 11 | 1,000   | 13     | <b>0.01300</b> | [0.0076, 0.0221] | <b>493</b>     | 0.03x  |
| PyMatching v2 (C++)                   | 11 | 5,000   | 24     | <b>0.00480</b> | [0.0032, 0.0071] | <b>19,863</b>  | 1.00x  |
| QECTOR CUDA Batch (GPU, unweighted)   | 11 | 5,000   | 229    | <b>0.04580</b> | [0.0403, 0.0520] | <b>15,865</b>  | 0.80x  |
| QECTOR OpenCL Batch (GPU, unweighted) | 11 | 5,000   | 229    | <b>0.04580</b> | [0.0403, 0.0520] | <b>12,608</b>  | 0.64x  |
| QECTOR Union-Find (CPU)               | 11 | 5,000   | 71     | <b>0.01420</b> | [0.0113, 0.0179] | <b>957</b>     | 0.05x  |

| Decoder                               | d  | Shots   | Errors | LER            | 95% CI           | dec/s         | vs PM |
|---------------------------------------|----|---------|--------|----------------|------------------|---------------|-------|
| PyMatching v2 (C++)                   | 11 | 10,000  | 77     | <b>0.00770</b> | [0.0062, 0.0096] | <b>20,957</b> | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 11 | 10,000  | 440    | <b>0.04400</b> | [0.0402, 0.0482] | <b>13,405</b> | 0.64x |
| QECTOR OpenCL Batch (GPU, unweighted) | 11 | 10,000  | 440    | <b>0.04400</b> | [0.0402, 0.0482] | <b>10,709</b> | 0.51x |
| PyMatching v2 (C++)                   | 11 | 50,000  | 334    | <b>0.00668</b> | [0.0060, 0.0074] | <b>21,251</b> | 1.00x |
| QECTOR OpenCL Batch (GPU, unweighted) | 11 | 50,000  | 2111   | <b>0.04222</b> | [0.0405, 0.0440] | <b>8,153</b>  | 0.38x |
| PyMatching v2 (C++)                   | 11 | 100,000 | 647    | <b>0.00647</b> | [0.0060, 0.0070] | <b>21,244</b> | 1.00x |
| PyMatching v2 (C++)                   | 13 | 1,000   | 3      | <b>0.00300</b> | [0.0010, 0.0088] | <b>11,222</b> | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 13 | 1,000   | 39     | <b>0.03900</b> | [0.0287, 0.0529] | <b>3,219</b>  | 0.29x |
| QECTOR OpenCL Batch (GPU, unweighted) | 13 | 1,000   | 39     | <b>0.03900</b> | [0.0287, 0.0529] | <b>4,712</b>  | 0.42x |
| QECTOR Union-Find (CPU)               | 13 | 1,000   | 11     | <b>0.01100</b> | [0.0062, 0.0196] | <b>176</b>    | 0.02x |
| PyMatching v2 (C++)                   | 13 | 5,000   | 18     | <b>0.00360</b> | [0.0023, 0.0057] | <b>11,971</b> | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 13 | 5,000   | 201    | <b>0.04020</b> | [0.0351, 0.0460] | <b>9,106</b>  | 0.76x |
| QECTOR OpenCL Batch (GPU, unweighted) | 13 | 5,000   | 201    | <b>0.04020</b> | [0.0351, 0.0460] | <b>7,145</b>  | 0.60x |
| PyMatching v2 (C++)                   | 13 | 10,000  | 37     | <b>0.00370</b> | [0.0027, 0.0051] | <b>12,135</b> | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 13 | 10,000  | 410    | <b>0.04100</b> | [0.0373, 0.0451] | <b>7,612</b>  | 0.63x |
| QECTOR OpenCL Batch (GPU, unweighted) | 13 | 10,000  | 410    | <b>0.04100</b> | [0.0373, 0.0451] | <b>4,590</b>  | 0.38x |
| PyMatching v2 (C++)                   | 13 | 50,000  | 221    | <b>0.00442</b> | [0.0039, 0.0050] | <b>11,935</b> | 1.00x |
| PyMatching v2 (C++)                   | 13 | 100,000 | 445    | <b>0.00445</b> | [0.0041, 0.0049] | <b>11,931</b> | 1.00x |
| PyMatching v2 (C++)                   | 15 | 1,000   | 2      | <b>0.00200</b> | [0.0005, 0.0073] | <b>5,968</b>  | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 15 | 1,000   | 41     | <b>0.04100</b> | [0.0304, 0.0551] | <b>1,929</b>  | 0.32x |
| QECTOR OpenCL Batch (GPU, unweighted) | 15 | 1,000   | 41     | <b>0.04100</b> | [0.0304, 0.0551] | <b>2,775</b>  | 0.47x |
| PyMatching v2 (C++)                   | 15 | 5,000   | 12     | <b>0.00240</b> | [0.0014, 0.0042] | <b>6,814</b>  | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 15 | 5,000   | 170    | <b>0.03400</b> | [0.0293, 0.0394] | <b>5,347</b>  | 0.79x |
| QECTOR OpenCL Batch (GPU, unweighted) | 15 | 5,000   | 170    | <b>0.03400</b> | [0.0293, 0.0394] | <b>2,662</b>  | 0.39x |
| PyMatching v2 (C++)                   | 15 | 10,000  | 44     | <b>0.00440</b> | [0.0033, 0.0059] | <b>7,061</b>  | 1.00x |
| QECTOR CUDA Batch (GPU, unweighted)   | 15 | 10,000  | 376    | <b>0.03760</b> | [0.0340, 0.0415] | <b>4,486</b>  | 0.64x |
| QECTOR OpenCL Batch (GPU, unweighted) | 15 | 10,000  | 376    | <b>0.03760</b> | [0.0340, 0.0415] | <b>2,798</b>  | 0.40x |
| PyMatching v2 (C++)                   | 15 | 50,000  | 171    | <b>0.00342</b> | [0.0029, 0.0040] | <b>7,240</b>  | 1.00x |
| PyMatching v2 (C++)                   | 15 | 100,000 | 314    | <b>0.00314</b> | [0.0028, 0.0035] | <b>7,296</b>  | 1.00x |

## Not measured (73 cells)

Listed so the gaps are explicit. These cells were never run and carry no numbers; nothing here was extrapolated or synthesised.

| Decoder                               | d  | Shots   | Reason                      | Projected |
|---------------------------------------|----|---------|-----------------------------|-----------|
| ldpc BP-OSD                           | 3  | 100,000 | over per-cell decode budget | 40s       |
| ldpc BP-OSD                           | 5  | 5,000   | over per-cell decode budget | 47s       |
| ldpc BP-OSD                           | 5  | 10,000  | over per-cell decode budget | 94s       |
| ldpc BP-OSD                           | 5  | 50,000  | over per-cell decode budget | 470s      |
| ldpc BP-OSD                           | 5  | 100,000 | over per-cell decode budget | 940s      |
| QECTOR Sparse Blossom (CPU)           | 7  | 50,000  | over per-cell decode budget | 31s       |
| QECTOR Sparse Blossom (CPU)           | 7  | 100,000 | over per-cell decode budget | 61s       |
| ldpc BP-OSD                           | 7  | 1,000   | over per-cell decode budget | 36s       |
| ldpc BP-OSD                           | 7  | 5,000   | over per-cell decode budget | 179s      |
| ldpc BP-OSD                           | 7  | 10,000  | over per-cell decode budget | 357s      |
| ldpc BP-OSD                           | 7  | 50,000  | over per-cell decode budget | 1,785s    |
| ldpc BP-OSD                           | 7  | 100,000 | over per-cell decode budget | 3,570s    |
| QECTOR Sparse Blossom (CPU)           | 9  | 10,000  | over per-cell decode budget | 45s       |
| QECTOR Sparse Blossom (CPU)           | 9  | 50,000  | over per-cell decode budget | 224s      |
| QECTOR Sparse Blossom (CPU)           | 9  | 100,000 | over per-cell decode budget | 448s      |
| QECTOR CUDA Batch (GPU, unweighted)   | 9  | 100,000 | over per-cell decode budget | 32s       |
| ldpc BP-OSD                           | 9  | 1,000   | over per-cell decode budget | 93s       |
| ldpc BP-OSD                           | 9  | 5,000   | over per-cell decode budget | 463s      |
| ldpc BP-OSD                           | 9  | 10,000  | over per-cell decode budget | 925s      |
| ldpc BP-OSD                           | 9  | 50,000  | over per-cell decode budget | 4,627s    |
| ldpc BP-OSD                           | 9  | 100,000 | over per-cell decode budget | 9,255s    |
| QECTOR Sparse Blossom (CPU)           | 11 | 5,000   | over per-cell decode budget | 64s       |
| QECTOR Sparse Blossom (CPU)           | 11 | 10,000  | over per-cell decode budget | 129s      |
| QECTOR Sparse Blossom (CPU)           | 11 | 50,000  | over per-cell decode budget | 644s      |
| QECTOR Sparse Blossom (CPU)           | 11 | 100,000 | over per-cell decode budget | 1,288s    |
| QECTOR Union-Find (CPU)               | 11 | 10,000  | over per-cell decode budget | 40s       |
| QECTOR Union-Find (CPU)               | 11 | 50,000  | over per-cell decode budget | 199s      |
| QECTOR Union-Find (CPU)               | 11 | 100,000 | over per-cell decode budget | 398s      |
| QECTOR CUDA Batch (GPU, unweighted)   | 11 | 50,000  | over per-cell decode budget | 34s       |
| QECTOR CUDA Batch (GPU, unweighted)   | 11 | 100,000 | over per-cell decode budget | 68s       |
| QECTOR OpenCL Batch (GPU, unweighted) | 11 | 100,000 | over per-cell decode budget | 37s       |
| ldpc BP-OSD                           | 11 | 1,000   | over per-cell decode budget | 202s      |
| ldpc BP-OSD                           | 11 | 5,000   | over per-cell decode budget | 1,009s    |
| ldpc BP-OSD                           | 11 | 10,000  | over per-cell decode budget | 2,018s    |
| ldpc BP-OSD                           | 11 | 50,000  | over per-cell decode budget | 10,092s   |
| ldpc BP-OSD                           | 11 | 100,000 | over per-cell decode budget | 20,184s   |
| QECTOR Sparse Blossom (CPU)           | 13 | 1,000   | over per-cell decode budget | 33s       |
| QECTOR Sparse Blossom (CPU)           | 13 | 5,000   | over per-cell decode budget | 166s      |
| QECTOR Sparse Blossom (CPU)           | 13 | 10,000  | over per-cell decode budget | 331s      |
| QECTOR Sparse Blossom (CPU)           | 13 | 50,000  | over per-cell decode budget | 1,656s    |
| QECTOR Sparse Blossom (CPU)           | 13 | 100,000 | over per-cell decode budget | 3,311s    |
| QECTOR Union-Find (CPU)               | 13 | 5,000   | over per-cell decode budget | 55s       |

| Decoder                               | d  | Shots   | Reason                      | Projected |
|---------------------------------------|----|---------|-----------------------------|-----------|
| QECTOR Union-Find (CPU)               | 13 | 10,000  | over per-cell decode budget | 111s      |
| QECTOR Union-Find (CPU)               | 13 | 50,000  | over per-cell decode budget | 554s      |
| QECTOR Union-Find (CPU)               | 13 | 100,000 | over per-cell decode budget | 1,107s    |
| QECTOR CUDA Batch (GPU, unweighted)   | 13 | 50,000  | over per-cell decode budget | 58s       |
| QECTOR CUDA Batch (GPU, unweighted)   | 13 | 100,000 | over per-cell decode budget | 116s      |
| QECTOR OpenCL Batch (GPU, unweighted) | 13 | 50,000  | over per-cell decode budget | 33s       |
| QECTOR OpenCL Batch (GPU, unweighted) | 13 | 100,000 | over per-cell decode budget | 67s       |
| ldpc BP-OSD                           | 13 | 1,000   | over per-cell decode budget | 341s      |
| ldpc BP-OSD                           | 13 | 5,000   | over per-cell decode budget | 1,707s    |
| ldpc BP-OSD                           | 13 | 10,000  | over per-cell decode budget | 3,413s    |
| ldpc BP-OSD                           | 13 | 50,000  | over per-cell decode budget | 17,065s   |
| ldpc BP-OSD                           | 13 | 100,000 | over per-cell decode budget | 34,130s   |
| QECTOR Sparse Blossom (CPU)           | 15 | 1,000   | over per-cell decode budget | 82s       |
| QECTOR Sparse Blossom (CPU)           | 15 | 5,000   | over per-cell decode budget | 412s      |
| QECTOR Sparse Blossom (CPU)           | 15 | 10,000  | over per-cell decode budget | 824s      |
| QECTOR Sparse Blossom (CPU)           | 15 | 50,000  | over per-cell decode budget | 4,120s    |
| QECTOR Sparse Blossom (CPU)           | 15 | 100,000 | over per-cell decode budget | 8,240s    |
| QECTOR Union-Find (CPU)               | 15 | 1,000   | over per-cell decode budget | 30s       |
| QECTOR Union-Find (CPU)               | 15 | 5,000   | over per-cell decode budget | 152s      |
| QECTOR Union-Find (CPU)               | 15 | 10,000  | over per-cell decode budget | 303s      |
| QECTOR Union-Find (CPU)               | 15 | 50,000  | over per-cell decode budget | 1,517s    |
| QECTOR Union-Find (CPU)               | 15 | 100,000 | over per-cell decode budget | 3,034s    |
| QECTOR CUDA Batch (GPU, unweighted)   | 15 | 50,000  | over per-cell decode budget | 100s      |
| QECTOR CUDA Batch (GPU, unweighted)   | 15 | 100,000 | over per-cell decode budget | 200s      |
| QECTOR OpenCL Batch (GPU, unweighted) | 15 | 50,000  | over per-cell decode budget | 60s       |
| QECTOR OpenCL Batch (GPU, unweighted) | 15 | 100,000 | over per-cell decode budget | 121s      |
| ldpc BP-OSD                           | 15 | 1,000   | over per-cell decode budget | 548s      |
| ldpc BP-OSD                           | 15 | 5,000   | over per-cell decode budget | 2,738s    |
| ldpc BP-OSD                           | 15 | 10,000  | over per-cell decode budget | 5,475s    |
| ldpc BP-OSD                           | 15 | 50,000  | over per-cell decode budget | 27,377s   |
| ldpc BP-OSD                           | 15 | 100,000 | over per-cell decode budget | 54,753s   |